

AI in Real World Context

A 60-minute field guide to tools, services and next-token predictors

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Contents

▷ 1. Prologue: AI in the Real World

- Real → Current
- The Question

▷ 2. The Agenda

- Reading the Room
- The Goal

▷ 3. Tool Time

- What Kind of Tools?
- Are All These AI Tools?

▷ 4. The AI Landscape

- The Umbrella Term
- Next-Token Predictors
- The Lens & The Assistant Axis
- The Alignment Wall

▷ 5. Fundamentals

- Services
- Side Note
- Where Information Lives
- It's All Text

▷ 6. Knowledge Compression

- Human Knowledge → Model
- The Distribution
- Too Much Maths?

▷ 7. Tools You Should Learn

- Git & Friends
- L^AT_EX & Overleaf
- Do It Yourself?

▷ 8. Agentic Tools

- Use Agentic Tools
- The Definition

1. Prologue: AI in the Real World

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.
- ◇ **Current**: what exists *today*, in your browser, your phone, your lab.

AI in Real → Current World Context

- ◇ **Real** promises a fixed world — it isn't one.
- ◇ **Current**: what exists *today*, in your browser, your phone, your lab.
- ◇ Everything in this talk is something you can touch **this week**.

so, let's start with you

What do you mean by “AI”?

◇ ChatGPT / Gemini / Copilot?

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- ◇ Pure hype?

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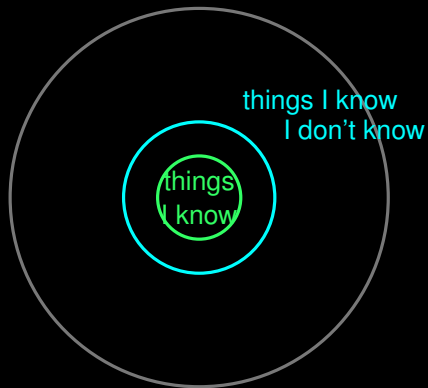
the deal

Whatever you said — we'll build on it; by the end you'll have a sharper answer.

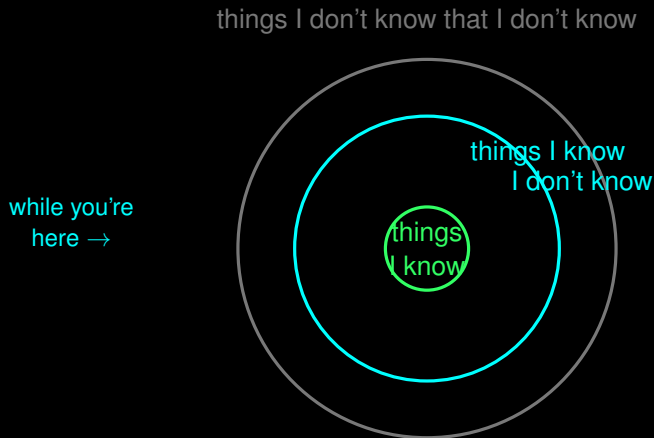
2. The Agenda

The room, mapped | three circles of knowing

things I don't know that I don't know



The room, mapped | three circles of knowing



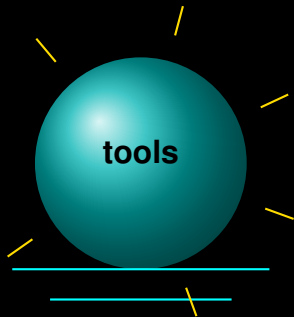
Objective: increase the area of **things you know you don't know**.

The room, mapped | three circles of knowing



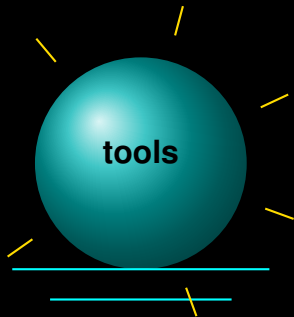
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I had a premonition | and it was about tools



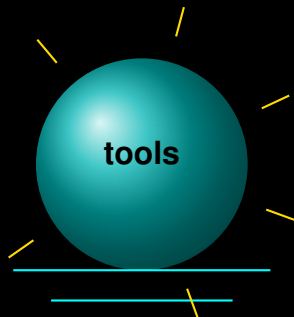
- ◇ The audience wants **tools**, not theory.

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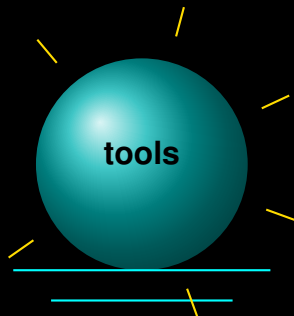
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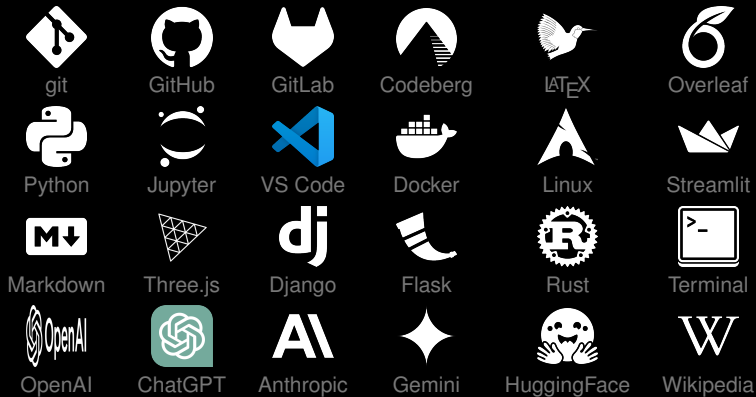
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Goal #1

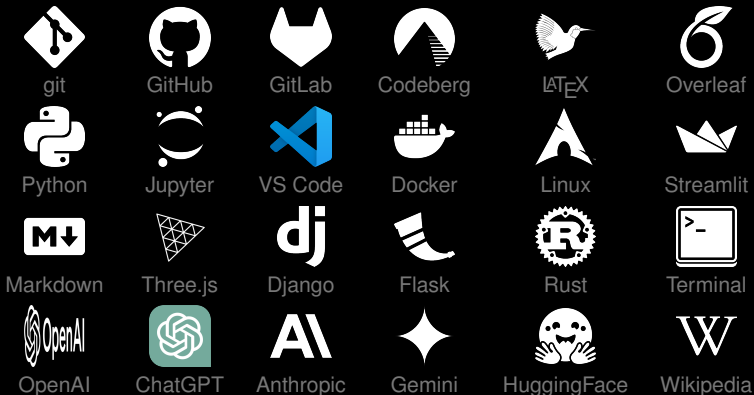
In the next **1 hour**, you will learn about tools.

3. Tool Time

What kind of tools? | spoiler: not all of these are AI



What kind of tools? | spoiler: not all of these are AI



Are all these AI tools?

Are all these AI tools? | nope — and that's exactly the point

Most of them aren't AI.

Git, $\text{\LaTeX}$, the terminal — older than most of you, zero AI.
The AI ones (ChatGPT, Gemini, Claude) *sit on top* of these.

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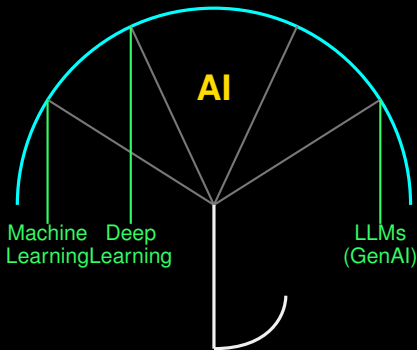
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the point

The non-AI tools are **leverage**: learn them once, use them forever.
With them, the AI tools become **superpowers** instead of toys.

4. The AI Landscape

AI is an umbrella term | what I work in: LLMs

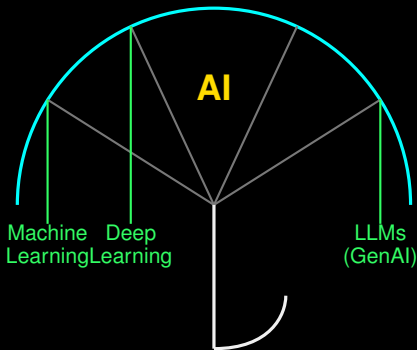


- ◇ **AI** = the umbrella over everything "intelligent".



xkcd 1838 — machine learning, the honest version

AI is an umbrella term | what I work in: LLMs



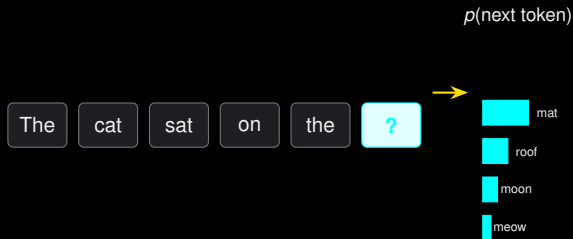
- ◇ **AI** = the umbrella over everything "intelligent".
- ◇ What I work in: **LLMs** — *next-token predictors*.



xkcd 1838 — machine learning, the honest version

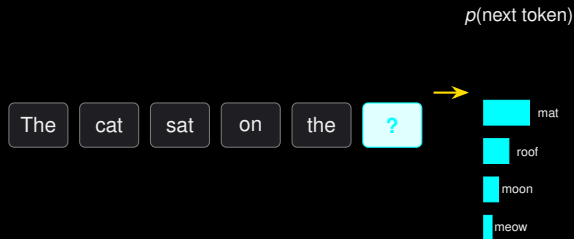
LLMs = next-token predictors | everything ChatGPT does, one loop

- ◇ The model never *searches* — it *predicts*.



Input: a token sequence $\longrightarrow$ **Output:** probabilities over the next token.

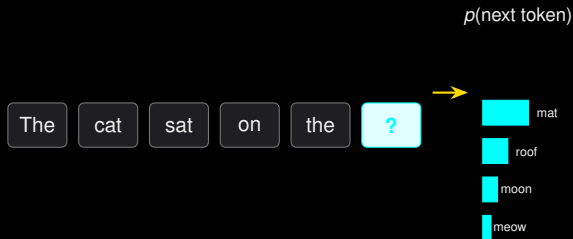
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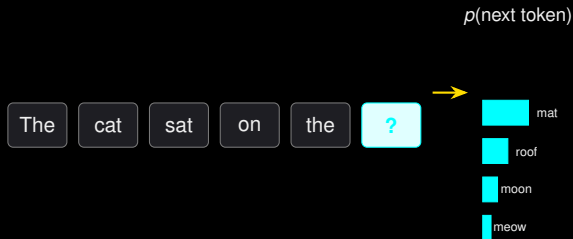
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trust me bro — I know this token

Through the lens | once you see it, you can't unsee it

deep learning AGI reasoning
chatbots agents type

**next token
prediction**

$$p(w_t \mid w_{<t})$$

- ◇ The **lens**: look at any AI through it.

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- ◇ **Alignment**? steering the same predictor.

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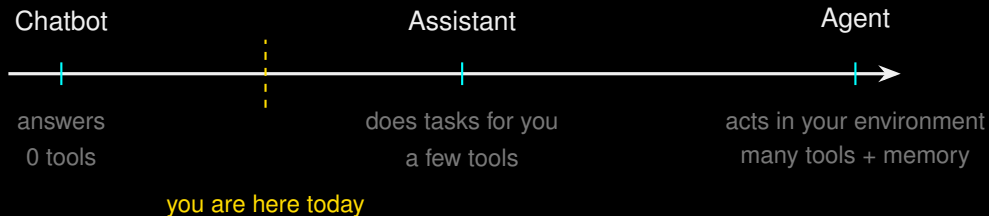
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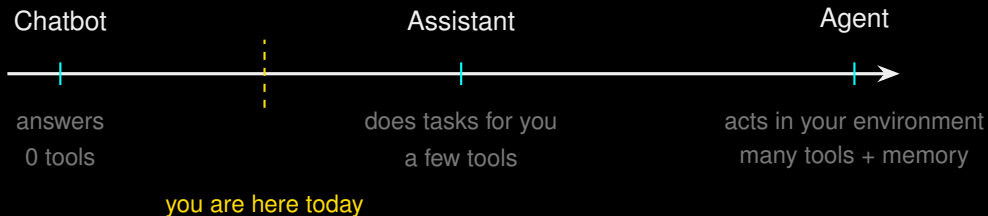
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- ◇ The **lens**: look at any AI through it.
- ◇ **Agents**? next-token prediction + tools.
- ◇ **Alignment**? steering the same predictor.
- ◇ The hype is loud; the mechanism is simple.

The assistant axis | from chatbot to agent



The assistant axis | from chatbot to agent



Question: where do you want to live on this axis by the end of the year?

The alignment wall | a tiny corner of a huge research effort

Real research is happening on **AI alignment** — nine papers I keep going back to:

Concrete Problems in AI Safety

Amodei, Olah, et al.

2016 · [arXiv:1606.06565](#)

AI Safety via Debate

Irving, Christiano, Amodei

2018 · [arXiv:1805.00899](#)

Scalable Agent Alignment via Reward Modeling

Leike, Krueger, et al.

2018 · [arXiv:1811.07871](#)

Fine-Tuning LMs from Human Preferences

Ziegler, Stiennon, et al.

2019 · [arXiv:1909.08593](#)

Training a Helpful & Harmless Assistant (RLHF)

Bai, Jones, et al.

2022 · [arXiv:2204.05862](#)

Constitutional AI: Harmlessness from AI Feedback

Bai, Kadavath, et al.

2022 · [arXiv:2212.08073](#)

Towards Understanding Syco-phancy

Sharma, Tong, et al.

2023 · [arXiv:2310.13548](#)

Weak-to-Strong Generalization

Burns, Izmailov, et al.

2023 · [arXiv:2312.09390](#)

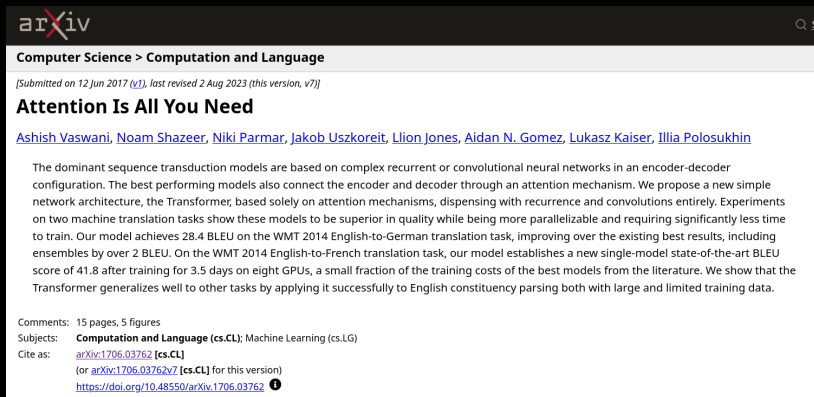
The AI Alignment Paradox

West & Aydin

2024 · [arXiv:2405.20806](#)

every paper is free on arXiv — pick one, read it tonight

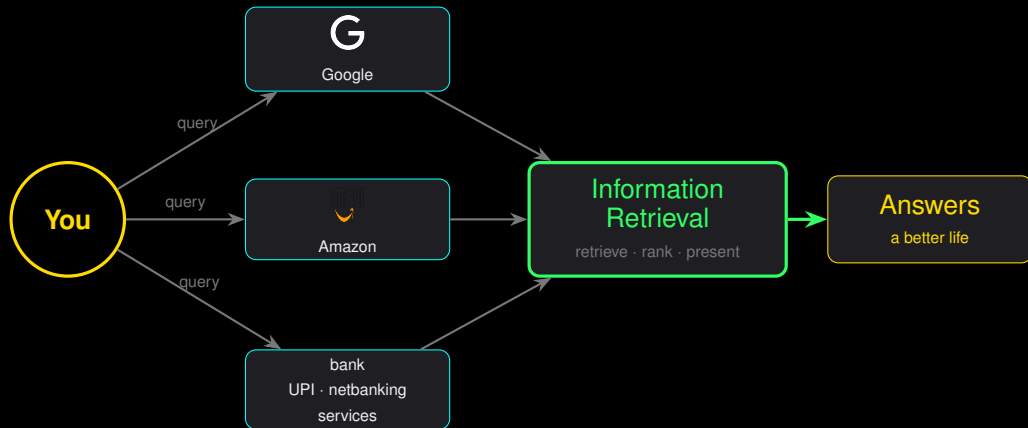
The paper that started this fourth wave | Attention Is All You Need



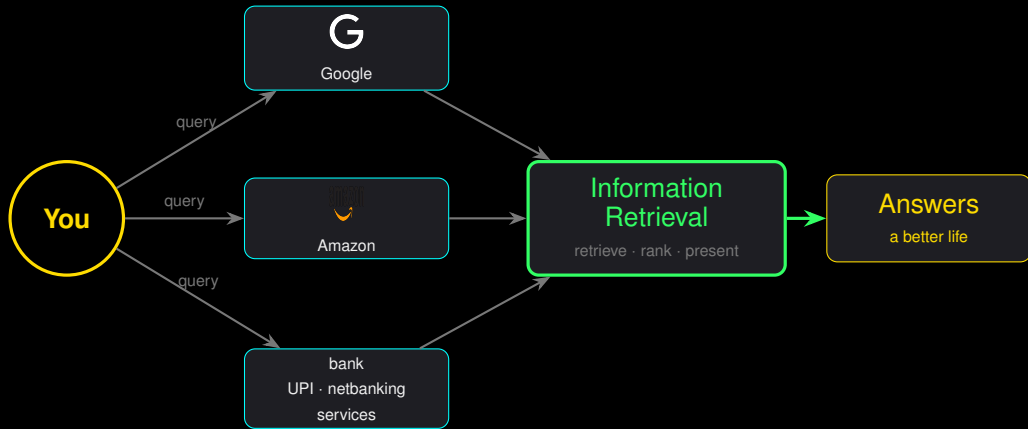
Vaswani et al., 2017 — **Attention Is All You Need**
arXiv:1706.03762

5. Fundamentals

How do you interact with computers? | spoiler: services



How do you interact with computers? | spoiler: services



Google, Amazon, even banks — they are **services**.
What do services do? **They retrieve information for you** — and make your life better.

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BS–MS final-year student, IISER Kolkata
Chemistry major · Computer Science minor
Research: **Information Retrieval** — RAG,
BM25, TREC, ECIR'26
Clubs: President, Slashdot · Office Bearer,
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- ◇ I study chemistry, I code for a living, I research **retrieval**.

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- ◇ I study chemistry, I code for a living, I research **retrieval**.
- ◇ I teach tools for fun — like this talk.
- ◇ Everything I show you, I use daily.

You tell me.

You tell me.

?



Video

most of the bytes



Photos

many bytes

W

Text

few bytes, most value

You tell me.

?



Video

most of the bytes



Photos

many bytes



Text

few bytes, most value

out of the three — **text** is the one that constitutes all these services and softwares.

It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?



It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?
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It's all text | the universal interface

- ◇ Out of **video**, **photos** and **text** — which one runs every service?
- ◇ **Text**. Text is the universal interface.
- ◇ Google? A text-index of the web. Your bank? Text rows in a database.



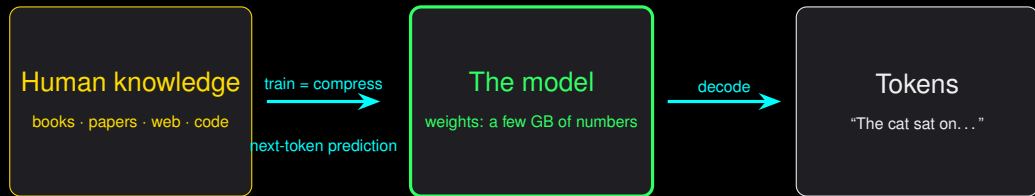
It's all text | the universal interface

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- ◇ Google? A text-index of the web. Your bank? Text rows in a database.
- ◇ LLMs eat text. **Agents act on text.**



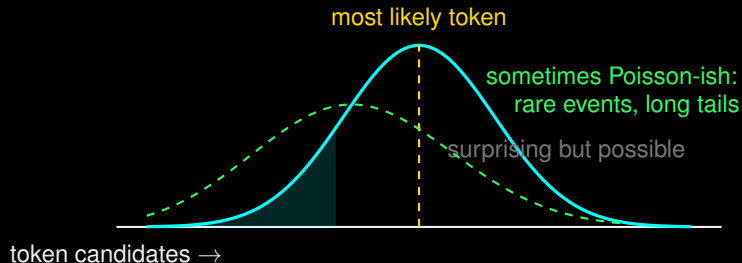
6. Knowledge Compression

LLMs = human knowledge, compressed into a model | the library in a file



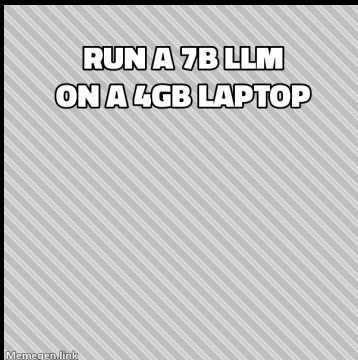
A whole library, compressed into one file — that predicts the next word.

The distribution of the next token | a Gaussian (or maybe a Poisson)



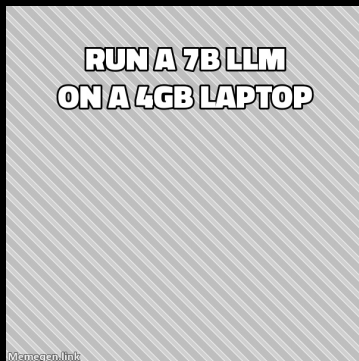
The model is a **probability distribution** over the next token — your knowledge, your language, your style, compressed.

Too much maths? | okay, that was the last formula



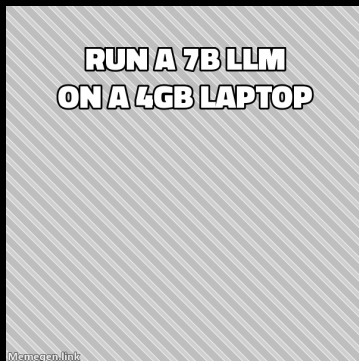
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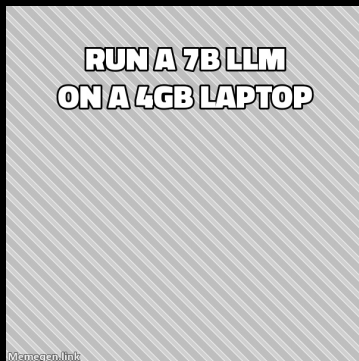
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- ◇ The takeaway is one line: **LLMs predict the next token.**
- ◇ And now that you know text matters. . .

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- ◇ Don't worry — that was the only formula in this talk.
- ◇ The takeaway is one line: **LLMs predict the next token.**
- ◇ And now that you know text matters. . .
- ◇ **let's get back to the tools.**

7. Tools You Should Learn

How many of you have a personal website? | hands up!

Hands up!

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Want one? Host it free on:



GitHub



GitLab



Codeberg

Underlying tool: **git**. You need to learn git.

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\$ your first repo

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$ git init
$ git add .
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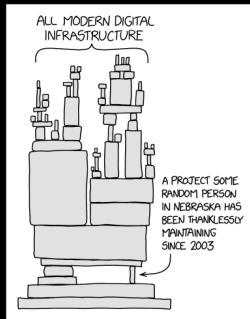


Codeberg

Underlying tool: **git**. You need to learn git.

\$ your first repo

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xkcd 2347: Dependency — everyone runs on git, eventually

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.



Overleaf



L^AT_EX



Overleaf



L^AT_EX

- ◇ PDFs: assignments, thesis, projects — all L^AT_EX.
- ◇ You type text + commands; the machine typesets beauty.



Overleaf



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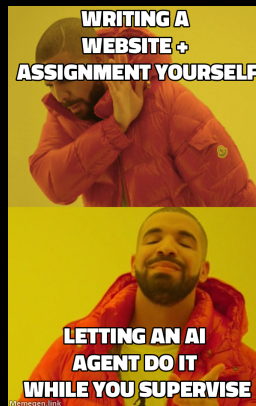
fun fact

Heck — even this presentation is made using L^AT_EX.
You are literally reading typeset tokens.

Would you do it by yourself? | now that everything is text

So now I've asked you to build a website and write an assignment.

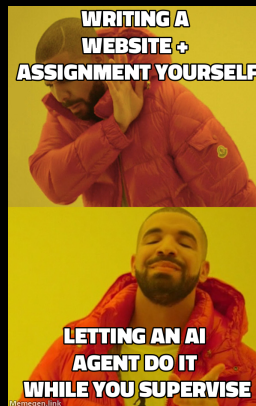
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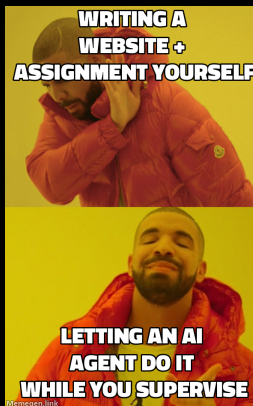
- ◇ Would you want to do it all **by yourself**?
- ◇ Remember: everything here is **text**.



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- ◇ Would you want to do it all **by yourself**?
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- ◇ And text is exactly what AI agents are good at.



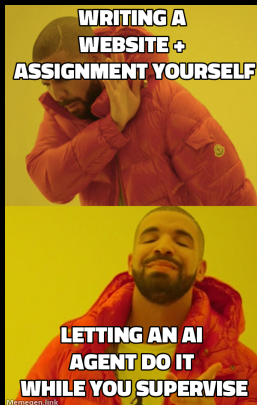
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the pitch

Use **agentic tools** — like the ones I'm about to show.



8. Agentic Tools

Use agentic tools | AI that acts, not just answers

- ◇ Tools I use daily: **VS Code** + **agentic coding tools**.

a session, roughly

```
$ opencode  
» write a python script that  
» downloads all my course PDFs  
✓ plan:  fetch, parse, save  
✓ wrote downloads.py  
✓ ran it:  42 files saved  
$
```

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- ◇ Open-source, runnable locally: OpenCode, Claude Code, ...

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OpenAI



ChatGPT



Claude



Gemini



HF



shell

a session, roughly

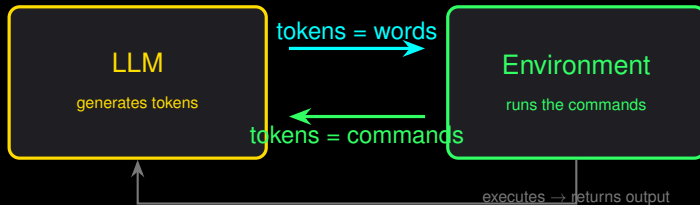
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What are agentic tools? | tokens become commands



What are agentic tools? | tokens become commands



LLMs generate tokens (words). What if the token is a **command** — and the environment it runs in **executes the command** as well? That's an **agentic tool**.

Questions? Everything is text.

Agents turn text into action.

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